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AI NASEEB

CBA0911

PSEUDO CODE

START

CLASS Student

Public name, Regno, marks[3]

Method Detail()

INPUT name

INPUT Reg

FOR i to n:

INPUT Marks[i]

END FOR

Method Average()

$$\text{Avg} = (\text{Mark}[1] + \text{Mark}[2] + \text{Mark}[3]) / 3$$

Return Avg

Method Display()

avg = Average()

Print "avg"

IF avg <= 50

PRINT "FAIL"

ELSE

PRINT "PASS"

END IF

Class main {

Create Student s

CALL s.Detail()

CALL s.display()

}

85/100

2. CLASS Book
Private title, author, ISBN, isAvailable.

Method getDetails ()

INPUT title, author, ISBN
is Available = True

Method Issuebook ()

If is Available = True

is Available = False

Print "Book Issued"

Else

Print "Not Avail"

Method returnbook ()

is available = TRUE

Print "Book Returned"

Method display ()

~~PRINT title, author, ISBN, isAvailable~~

Print get details ()

Class main {

Create book
~~Student~~ s

Call s.Details ()

Call s.display ()

}

START

CLASS Employee

Private empid,

Private name,

Private basic pay,

Private designation

Method Input ()

Input empid

Input name

Input designation

Input basic pay.

Method calculate ()

$HRA = \text{basic} * 0.20$

$DA = \text{basic} * 0.10$

$Gross = \text{basic} + HRA + DA$

$PF = \text{basic} * 0.12$

$Tax = Gross * 0.05$

$Net = Gross - PF - Tax$

~~Print Gross~~

~~Print Net~~

Method Display ()

PRINT NET

Class Main {

Create Employee . E1

Call methods

}

4.

START

CLASS SHAPE

METHOD calculate()

END CLASS

CLASS Circle extend Shape

OVERRIPE Method calculate()

Input radius

Area = $3.14 \times \text{radius} \times \text{radius}$

Print area.

CLASS Rectangle Extends Shape

OVER RIDE method calculate()

Input $\checkmark$ l, b

area = $l \times b$

PRINT area

Class ~~Rectangle~~ Triangle Extend Shape

Override method calculate()

Input l, b,

area = $l \times b \times 0.5$

PRINT area

Class Main $\uparrow$

Create ~~Shape~~ ^{Shape} b1.

Call methods

~

PART

class Vehicle

Method getter {

Int Rg

~~Int~~

String Name

String Fuel

}

Method Display() {

Print Rg, Name, Fuel }

END CLASS

CLASS TwoW Extends Vehicle

Override Method display()

Int Capacity.

Display()

CLASS FourW Extends Vehicle

Override Method Display()

Int Capacity

Display()

CLASS Main {

Create Vehicle .c

CALL Methods

}

6.

START

CLASS BANK ACC

Method getter 1/1

Int Acc NO

String Name

float Balance

}

Method withdraw 1/1

CLASS Savings Extends BankAccount.

Override method withdraw 1/1

Input Amt,

If Amt $\leq$ balance

Balance = Balance - Amt

Print "Balance"

else

"No, balance"

Class Overdraft Extends bankAccount.

Override method withdraw 1/1

Input Amt,

Class Main {

Create BankAcc . S

Call Method

}

Patient

```
method geller ( ) {
```

```
    String name
```

```
    int id
```

```
}
```

```
method generalbill ( )
```

```
CLASS Inpatient Extends Patient
```

```
    int Ward Number
```

```
    int Number of Days charges.
```

```
    int Days Days:
```

```
    Override method generalbill ( )
```

```
        Bill = Days x charges
```

```
CLASS Outpatient Extends Patient
```

```
    int Consultation Fee
```

```
    Override method generalbill ( )
```

```
        Bill = Consultation Fee
```

```
Class main {
```

```
    Create Patient . P
```

```
    CALL method
```

```
}
```

START

Class product

int price

String P_name

Method apply_Discount()

End method.

Class Electronics Extends product

Override Method applydiscount()

Discount = Price $\times$ 10 / 100

Final price = Price - Discount

PRINT "Final Price"

Class Clothing Extends product

Override method apply discount()

Discount = Price $\times$ 25 / 100

Final-price = Price - Discount

PRINT "FINAL Price"

Class main {

Create Product.P

CALL Method

}

Class Staff

Int Staff ID

String Name

Method calculate average()

End method.

Class Teaching Staff Extends Staffs

String Department

END CLASS

Class Professor Extends Teaching Staff
Override method calculate allowance()

Allowance = 20000

PRINT "NAME"

PRINT "Allowance"

Class ~~non~~Teaching Staff Extends Staff

~~Section~~ String Section

Override method calculate Allowance()

Allowance = 5000

PRINT "Staff name"

PRINT "Allowance"

Class Main {
Create Professor object
CALL calculate allowance()

}

START

Class Smart Device

String Name

Method execute Command()

CLASS SmartLight Extend Smart Devices

Override method execute Command()

Input Command

Print Device name

Print Command.

Class SmartThermometer Extends SmartDevices

Int Temperature

Override method execute Command()

Input Command

Print Name

Print Command

Class SmartLock Extends Smart device

Int Pin

Override Method execute Command()

Input Command.

Print Name

Print Command.

class main { End. class.

CREATE SmartLight object

CALL execute Command()